Laboratuvar Raporu 3

Eskişehir Osmangazi Üniversitesi

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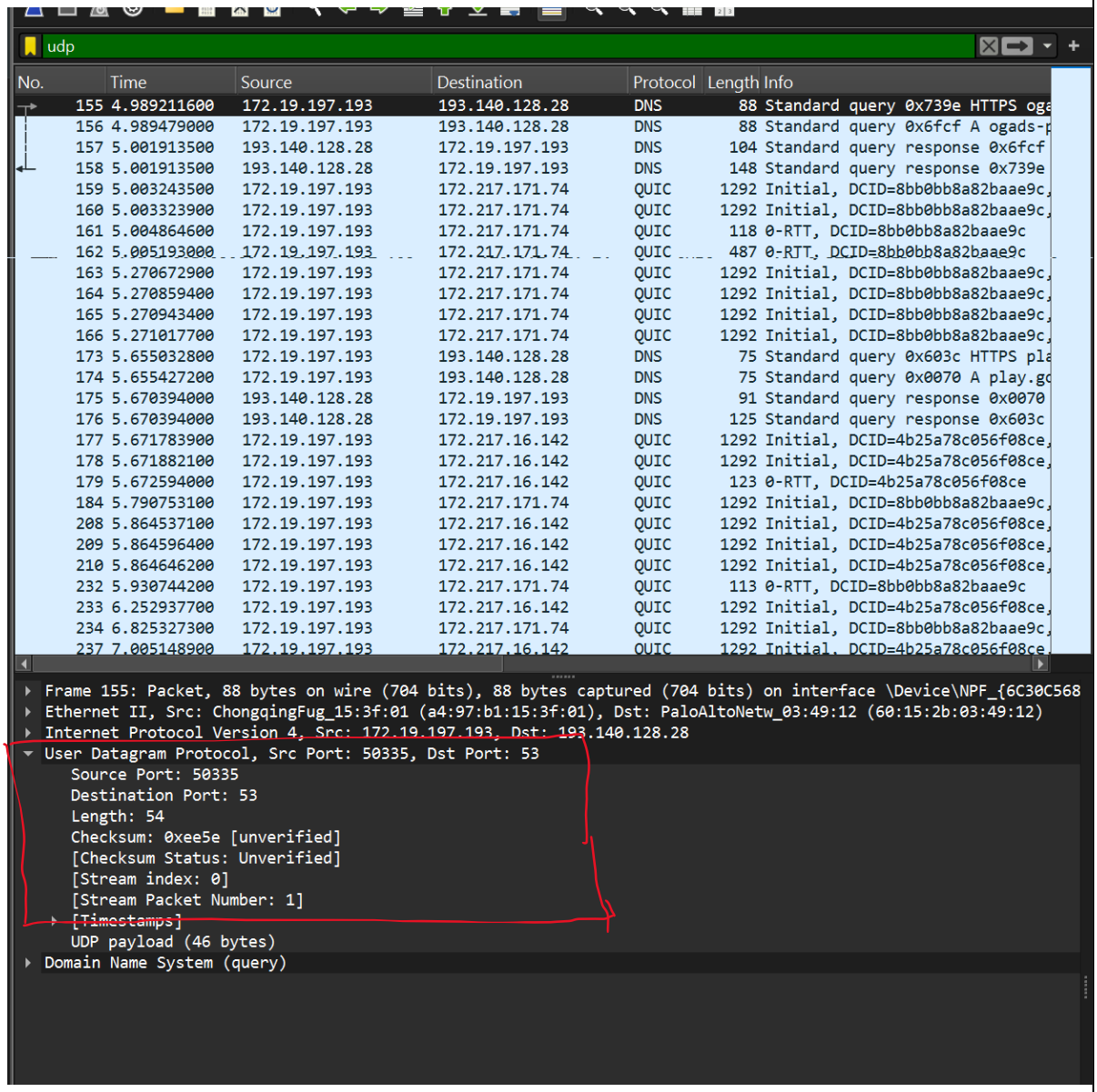
2025-2026

1 İçindekiler

2 Laboratuvar Uygulaması 3

2.Laboratuvar Uygulaması

3.1.



udp

No.	Time	Source	Destination	Protocol	Length	Info
155	4.989211600	172.19.197.193	193.140.128.28	DNS	88	Standard query 0x739e HTTPS oga
156	4.989479000	172.19.197.193	193.140.128.28	DNS	88	Standard query 0x6fcf A ogads-p
157	5.001913500	193.140.128.28	172.19.197.193	DNS	104	Standard query response 0x6fcf
158	5.001913500	193.140.128.28	172.19.197.193	DNS	148	Standard query response 0x739e
159	5.003243500	172.19.197.193	172.217.171.74	QUIC	1292	Initial, DCID=8bb0bb8a82baae9c,
160	5.003323900	172.19.197.193	172.217.171.74	QUIC	1292	Initial, DCID=8bb0bb8a82baae9c,
161	5.004864600	172.19.197.193	172.217.171.74	QUIC	118	0-RTT, DCID=8bb0bb8a82baae9c
162	5.005193000	172.19.197.193	172.217.171.74	QUIC	487	0-RTT, DCID=8bb0bb8a82baae9c
163	5.270672900	172.19.197.193	172.217.171.74	QUIC	1292	Initial, DCID=8bb0bb8a82baae9c,
164	5.270859400	172.19.197.193	172.217.171.74	QUIC	1292	Initial, DCID=8bb0bb8a82baae9c,
165	5.270943400	172.19.197.193	172.217.171.74	QUIC	1292	Initial, DCID=8bb0bb8a82baae9c,
166	5.271017700	172.19.197.193	172.217.171.74	QUIC	1292	Initial, DCID=8bb0bb8a82baae9c,
173	5.655032800	172.19.197.193	193.140.128.28	DNS	75	Standard query 0x603c HTTPS pla
174	5.655427200	172.19.197.193	193.140.128.28	DNS	75	Standard query 0x0070 A play.go
175	5.670394000	193.140.128.28	172.19.197.193	DNS	91	Standard query response 0x0070
176	5.670394000	193.140.128.28	172.19.197.193	DNS	125	Standard query response 0x603c
177	5.671783900	172.19.197.193	172.217.16.142	QUIC	1292	Initial, DCID=4b25a78c056f08ce,
178	5.671882100	172.19.197.193	172.217.16.142	QUIC	1292	Initial, DCID=4b25a78c056f08ce,
179	5.672594000	172.19.197.193	172.217.16.142	QUIC	123	0-RTT, DCID=4b25a78c056f08ce
184	5.790753100	172.19.197.193	172.217.171.74	QUIC	1292	Initial, DCID=8bb0bb8a82baae9c,
208	5.864537100	172.19.197.193	172.217.16.142	QUIC	1292	Initial, DCID=4b25a78c056f08ce,
209	5.864596400	172.19.197.193	172.217.16.142	QUIC	1292	Initial, DCID=4b25a78c056f08ce,
210	5.864646200	172.19.197.193	172.217.16.142	QUIC	1292	Initial, DCID=4b25a78c056f08ce,
232	5.930744200	172.19.197.193	172.217.171.74	QUIC	113	0-RTT, DCID=8bb0bb8a82baae9c
233	6.252937700	172.19.197.193	172.217.16.142	QUIC	1292	Initial, DCID=4b25a78c056f08ce,
234	6.825327300	172.19.197.193	172.217.171.74	QUIC	1292	Initial, DCID=8bb0bb8a82baae9c,
237	7.005148900	172.19.197.193	172.217.16.142	QUIC	1292	Initial, DCID=4b25a78c056f08ce,

```
Frame 155: Packet, 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface \Device\NPF_{6C30C568
Ethernet II, Src: ChongqingFug_15:3f:01 (a4:97:b1:15:3f:01), Dst: PaloAltoNetw_03:49:12 (60:15:2b:03:49:12)
Internet Protocol Version 4, Src: 172.19.197.193, Dst: 193.140.128.28
User Datagram Protocol, Src Port: 50335, Dst Port: 53
    Source Port: 50335
    Destination Port: 53
    Length: 54
    Checksum: 0xee5e [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 1]
    [Timestamps]
    UDP payload (46 bytes)
Domain Name System (query)
```

Toplam 4 adet alan:

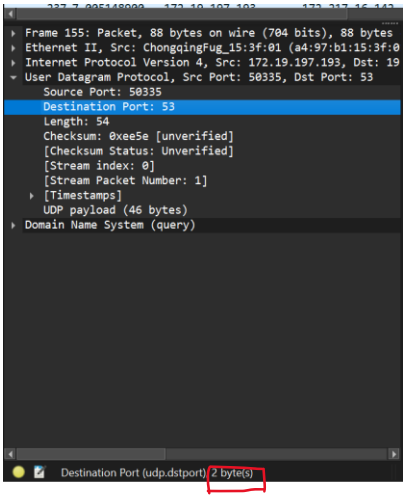
Source Port

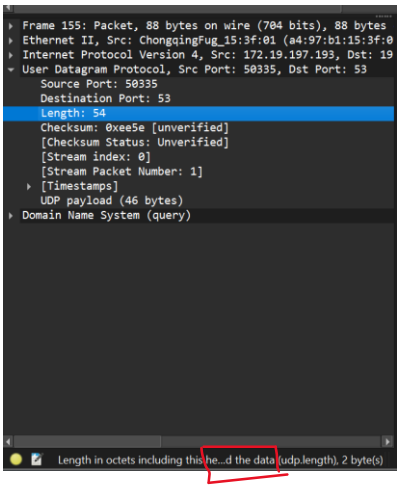
Destination Port

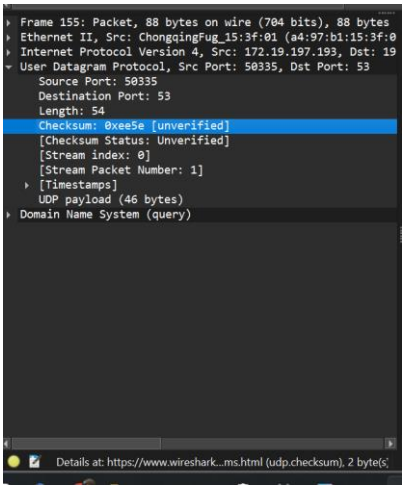
Length

Checksum

3.2.







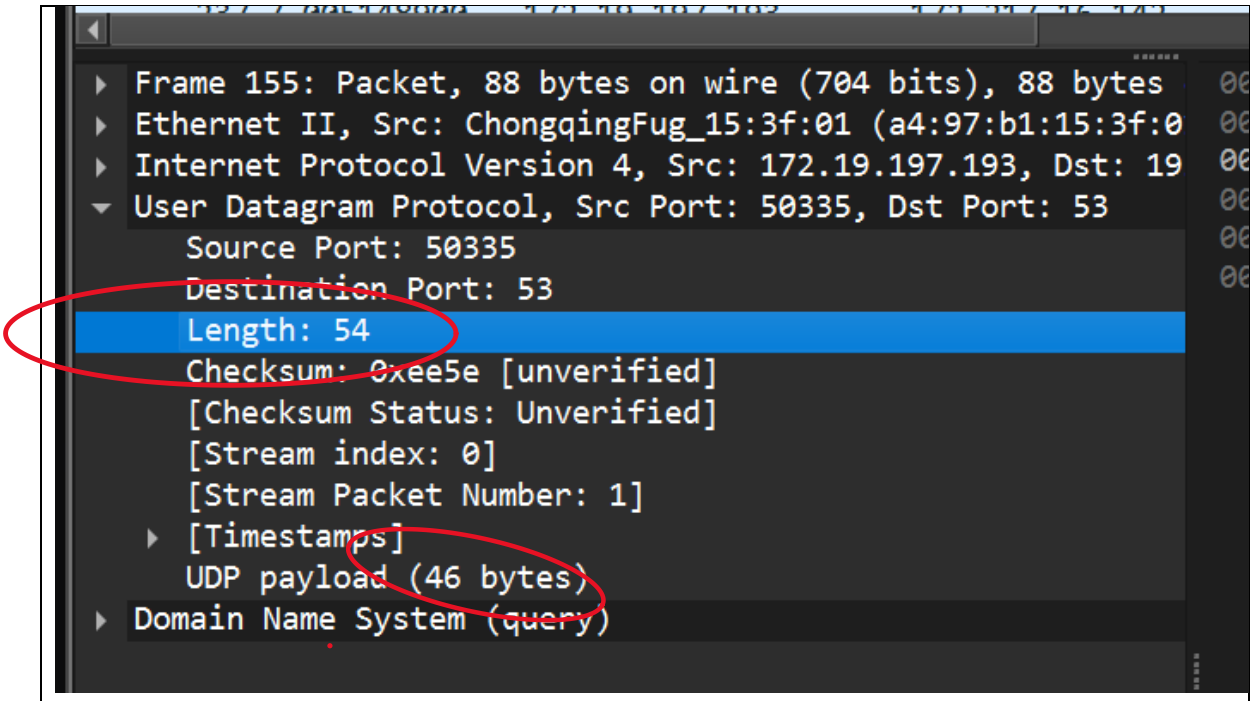
Source Port: 2 byte

Destination Port: 2 byte

Length: 2 byte

Checksum: 2 byte

3.3.



Length değeri 54 byte yani udp başlığı ve payloadın(46 byte) toplam uzunluğu.

3.4.

UDP başlığındaki Length alanı 2 bayt (16 bit) uzunluğundadır. 16 bit ile ifade edilebilecek en büyük sayı (2 üzeri 16) – 1 = 65535'dir.

Toplam Maksimum Uzunluk: 65.535 bayt .

UDP Başlık Boyutu: 8 bayt.

Maksimum Payload : 65.535 - 8 = 65.527 bayt.

Sonuç olarak, bir UDP paketinde taşınabilecek maksimum 65.527 bayt olabilir.

3.5.

```
Frame 155: Packet, 88 bytes on wire (704 bits), 88 bytes
Ethernet II, Src: ChongqingFug_15:3f:01 (a4:97:b1:15:3f:0
Internet Protocol Version 4, Src: 172.19.197.193, Dst: 19
User Datagram Protocol, Src Port: 50335, Dst Port: 53
    Source Port: 50335
    Destination Port: 53
    Length: 54
    Checksum: 0xee5e [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
    [Stream Packet Number: 1]
  ▸ [Timestamps]
    UDP payload (46 bytes)
Domain Name System (query)
```

```
0000  60 15 2b 03 49 12 a4 97  b1 15 3f 01 08 00 45
0010  00 4a 4f 08 00 00 80 11  38 1d ac 13 c5 c1 c1
0020  80 1c c4 9f 00 35 00 36  ee 5e 73 9e 01 00 00
0030  00 00 00 00 00 00 08 6f  67 61 64 73 2d 70 61
0040  63 6c 69 65 6e 74 73 36  06 67 6f 6f 67 6c 65
0050  63 6f 6d 00 00 41 00 01
```

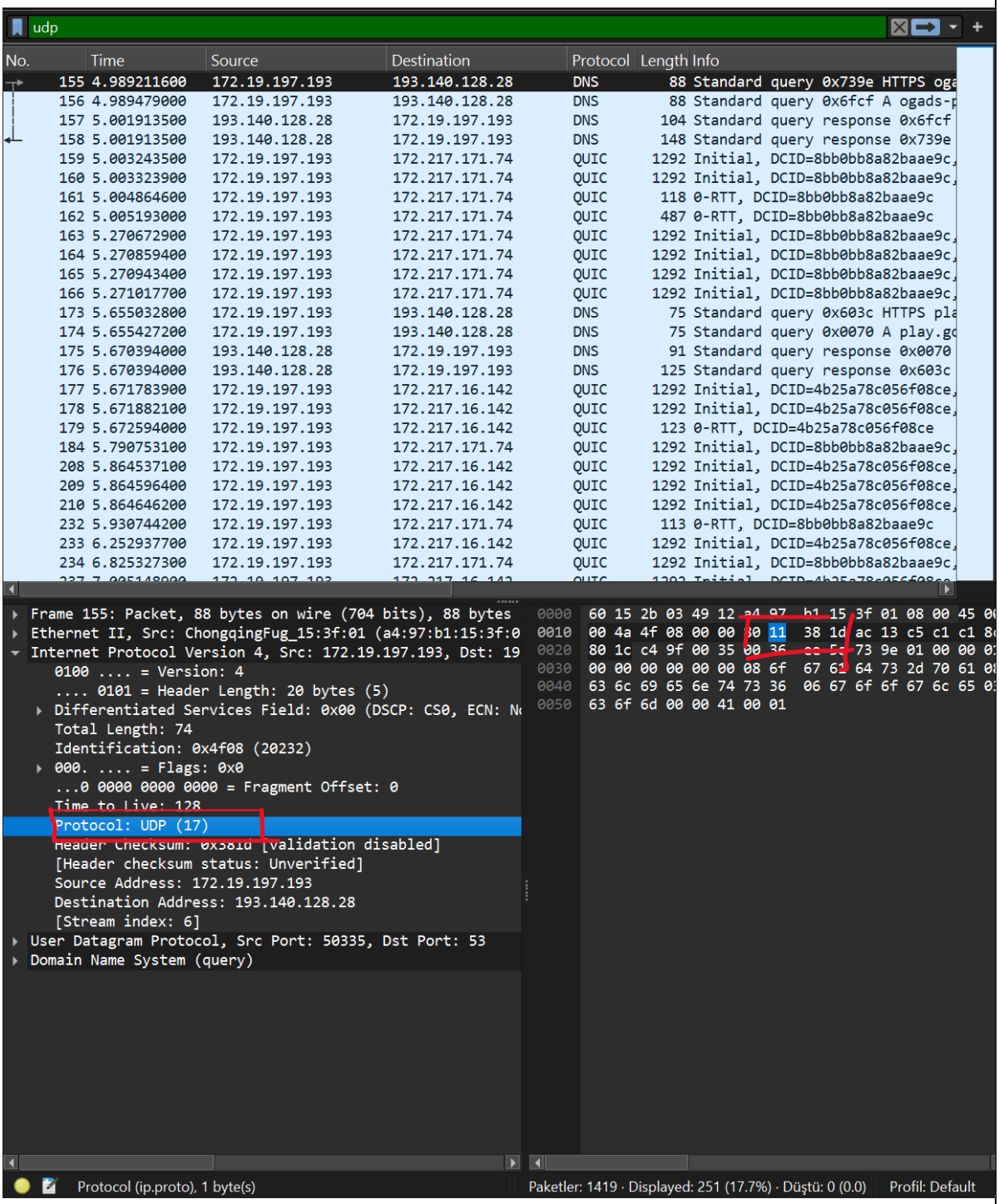
Source Port alanının 2 bayt (16 bit) uzunluğunda.

16 bitlik bir alanda tanımlanabilecek en büyük sayı (2^16) – 1 formülü ile hesaplanır.

Bu hesaplama sonucunda: 65.535 değerine ulaşılır.

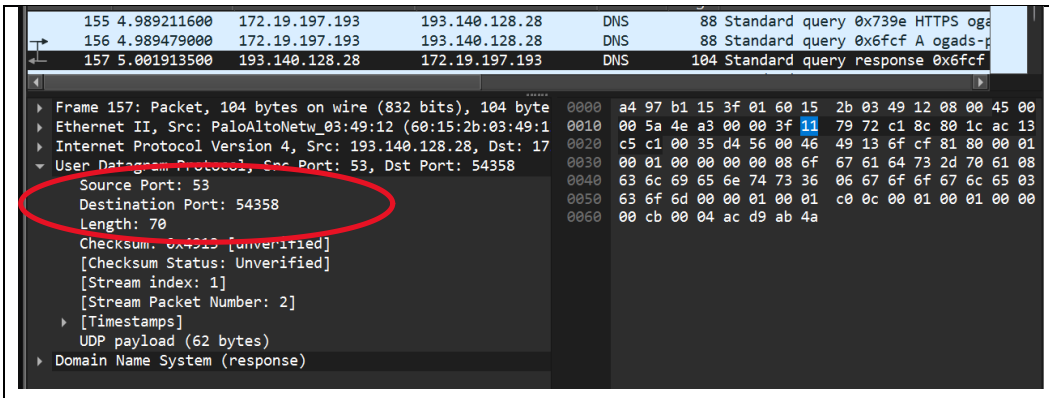
Bu UDP paketinde kullanılabilecek en büyük kaynak port numarası 65.535'tir.

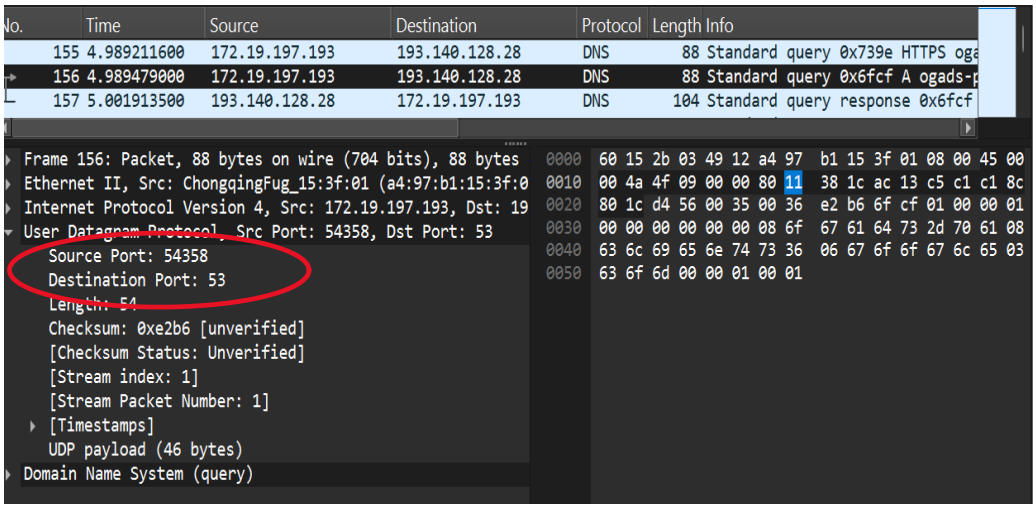
3.6.



Decimal olarak 17 hexadecimal olarak 11 değerine ulaşılmıştır.

3.7.





Frame 156 Source Port 54358 - Destination Port 53

Frame 157 Source Port 53 -Destination Port 54358

İstek paketindeki kaynak portu, cevap paketinde hedef portu olmuştur; istekteki hedef portu ise cevapta kaynak portu olarak kullanılmıştır. Bu port takası, sunucunun cevabı ağdaki doğru bilgisayara ve o bilgisayardaki doğru uygulamaya ulaştırmasını sağlar